



DALHOUSIE UNIVERSITY

CSCI 5308: Advanced Topics in Software Development

Assignment: 01

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Date: 2025-03-27

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1) Task 0

1. **Project name:** Jsoup
2. **Github Link:** <https://github.com/jhy/jsoup>
3. **Stars:** 11.1k
4. **Lines of code:** Above the 10,000 lines of code
5. **Language:** Java

I have fork in my Github account

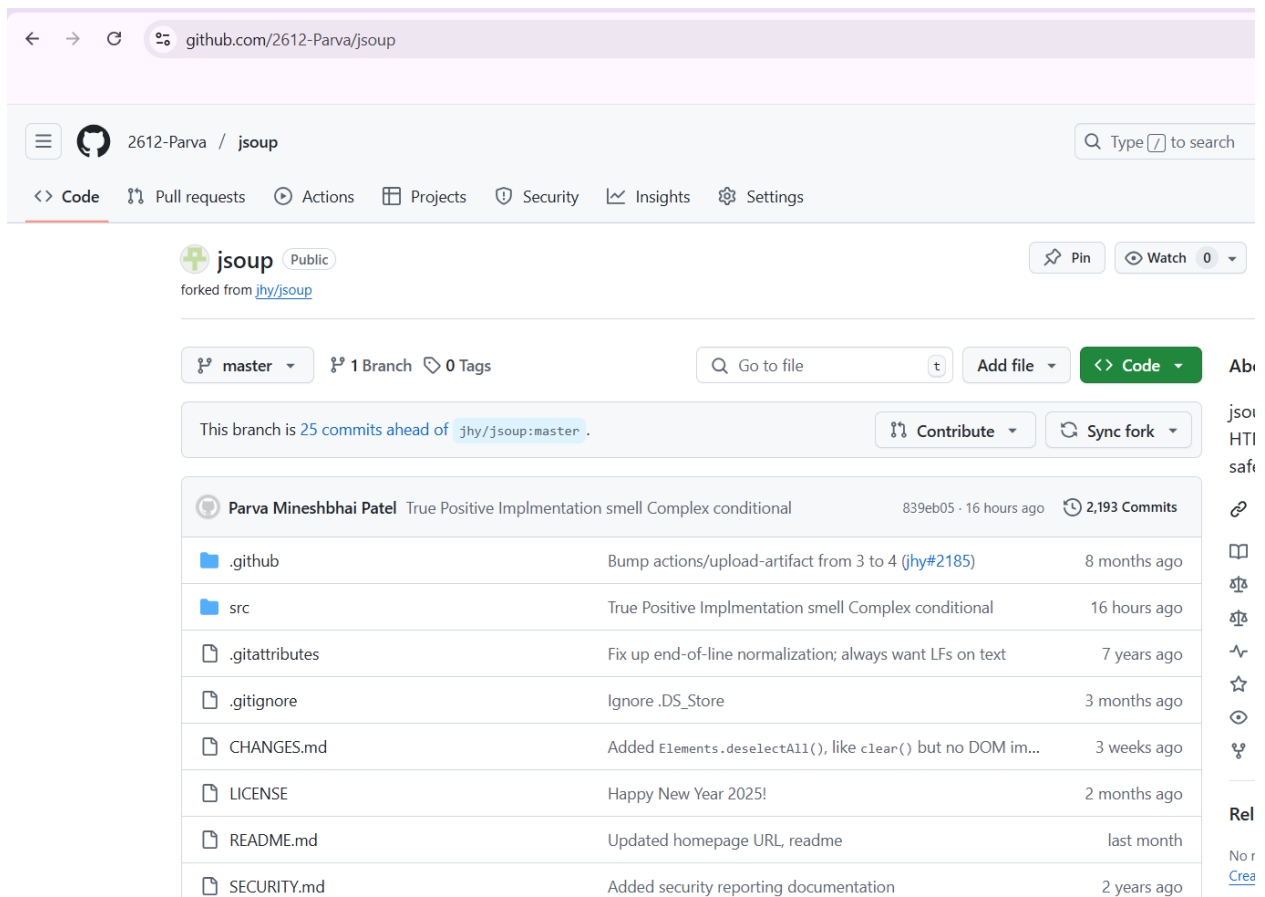


Figure 1: My Github Account

I have cloned the repository this is the link <https://github.com/2612-Parva/jsoup> using git clone command.

2) Task 1

Build completed successfully just after cloning

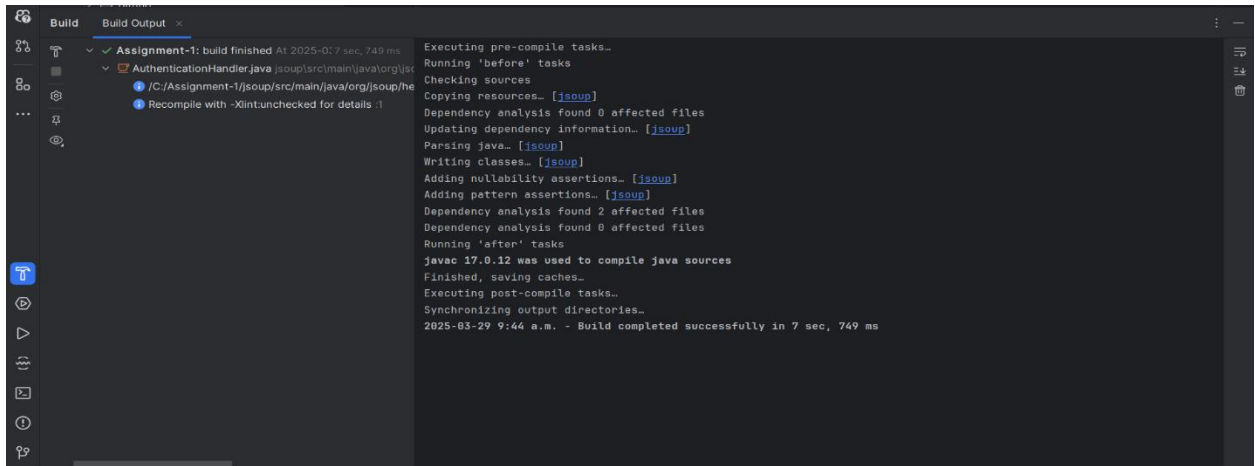


Figure 2: Build Successful

Set I

Implementation Smells

1) True Positive

1. Smell: Complex conditional

Location: src/main/java/org/jsoup/nodes/Attribute.java

Line number: 214

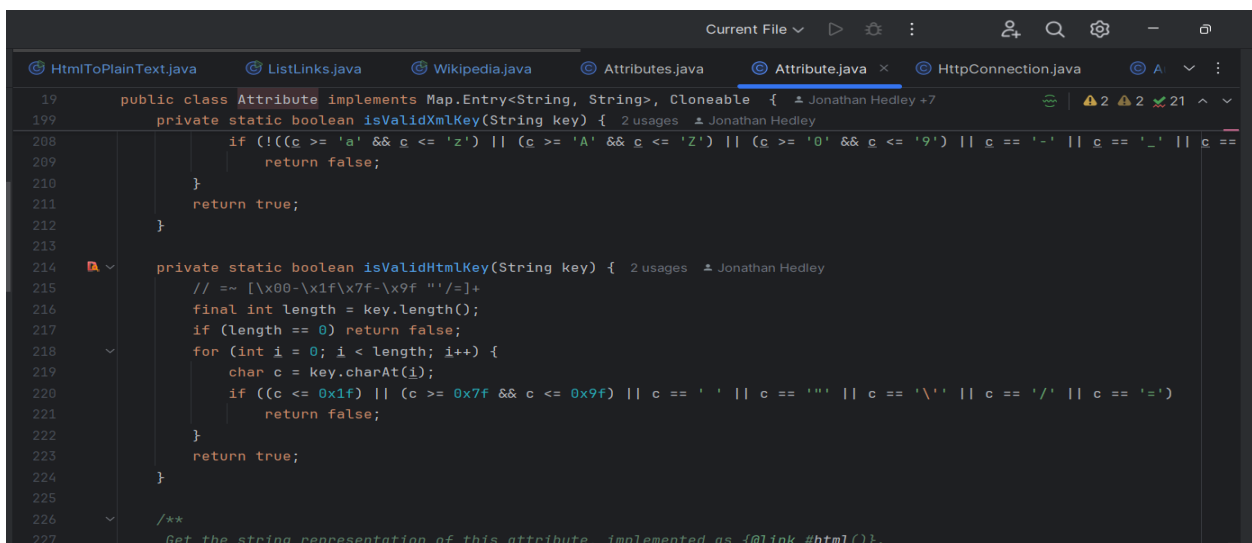


Figure 3: Complex Conditional smell(True Positive)

Implementation smell csv file

164	java	org.jsoup.nodes	Attribute	isValidHtmlKey	Complex Conditional
165	java	org.jsoup.nodes	Attribute	isValidHtmlKey	Magic Number

Figure 4: CSV file for Implementation smell

Explanation

In this `isValidHtmlKey` method, the conditional expression inside the `if` statement was complex due to multiple conditions being evaluated in a single statement. The expression included range comparisons (`c <= 0x1f`, `c >= 0x7f` && `c <= 0x9f`) along with multiple equality checks (`c == ' '`, `c == '\"'`, `c == '\\'`, `c == '/'`, `c == '='`). This resulted in reduced readability and maintainability. So, for this I will use the decompose conditional, Extract method and Introduce Explanation Variable refactoring technique. Thus, I agree with the tool that this is a true positive implementation smell categorized as a complex conditional.

2. Smell: Complex method and long statement

Location: `src/main/java/org/jsoup/helper/DataUtil.java`

Line number: 238

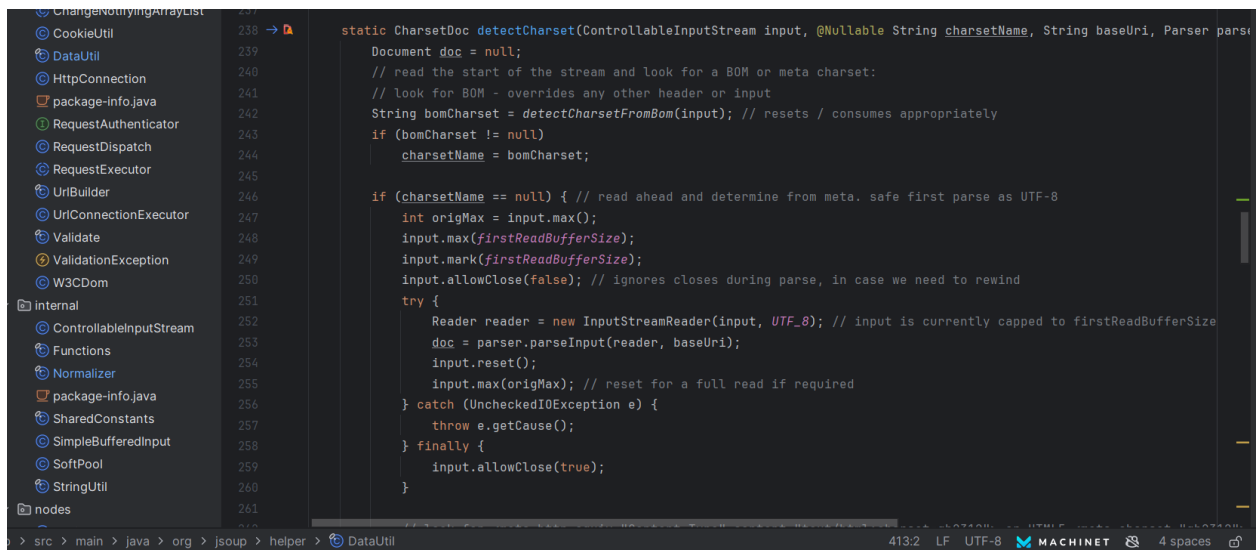


Figure 5: Complex method and long statement smell(True positive)

Implementation smell CSV file

17	java	org.jsoup.helper	DataUtil	detectCharsetFromBom	Complex Conditional
18	java	org.jsoup.helper	DataUtil	detectCharsetFromBom	Long Statement
19	java	org.jsoup.helper	DataUtil	detectCharsetFromBom	Magic Number

Figure 6: CSV file for Implementation smells

Explanation

This detectCharset method detects Complex Method and Long Statement implementation smells due to multiple responsibilities handled within a single method. It contained deeply nested conditionals, multiple decision points, and lengthy statements so it is reducing readability and maintainability. For that, I will use the Decompose Conditional, Introduce Explaining Variable and Extract Method refactor technique. So, I agree with the tool that this is the true positive implementation smell.

2) False Positive

1. Smell: Magic Number

Location: src/main/java/org/jsoup/examples/HtmlToPlainText.java

Line: 34

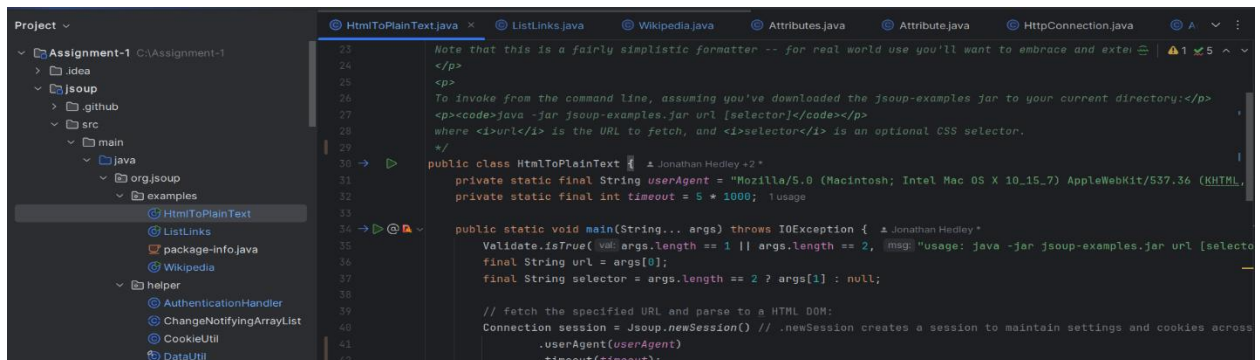


Figure 7: Magic Number Implementation smell(False Positive)

Implementation smell CSV file

	A	B	C	D	E
1	Project Na	Package Name	Type Name	Method Name	Code Smell
2	java	org.jsoup.examples	HtmlToPlainText	main	Magic Number

Figure 8: CSV file for Implementation smell

Explanation

The numbers 1 and 2 in this context are not arbitrary values but represent valid argument lengths required for proper execution of the program.

These values serve as explicit constraints for argument validation rather than unidentified or hardcoded magic numbers. Thus, this detection is a False Positive, and no refactoring is needed.

3) False Negative

1. Smell: Complex method

Location: src/main/java/org/jsoup/helper/AuthenticationHandler.java

Line: 14

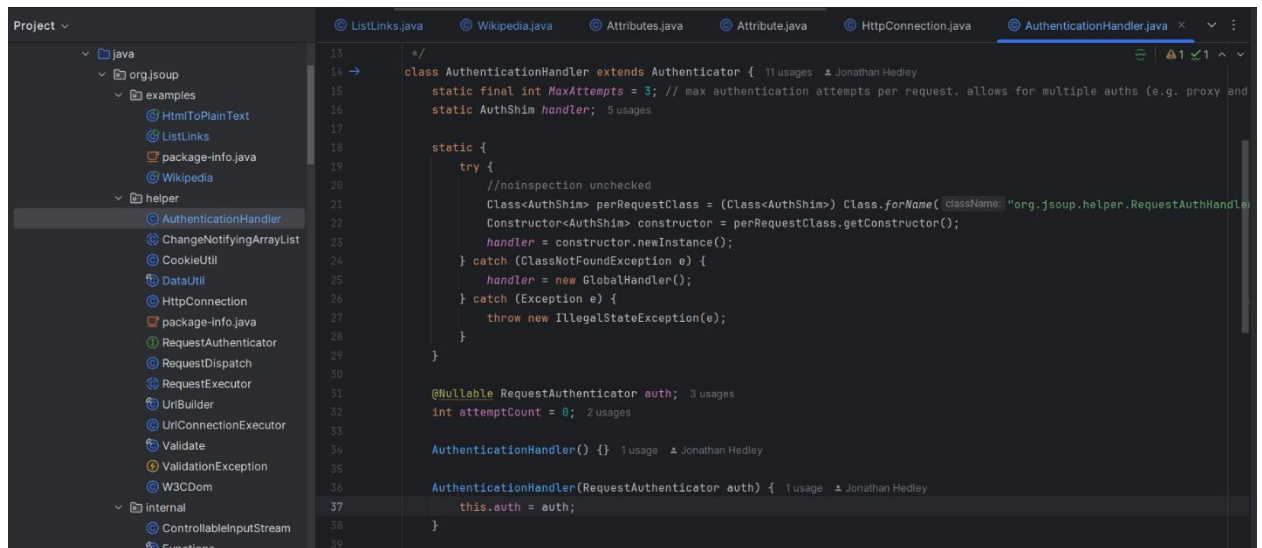


Figure 9: Complex Method Implementation Smell(False Negative)

Explanation

The tool failed to detect a False Negative for the Complex Method implementation smell in the getPasswordAuthentication() method. This method handled multiple responsibilities, including retrieving the authentication delegate, managing authentication attempt limits, checking whether authentication should proceed, creating an authentication context, and returning credentials. So, I think this method is more complex and harder to read. The tool failed to recognize this complexity so this is classified as a False Negative an undetected issue that was manually identified.

Set II

Design Smells

1) True Positive

1. Smell: Unnecessary Abstraction

Location: src/main/java/org/jsoup/internal/StringUtil.java

Line: 71

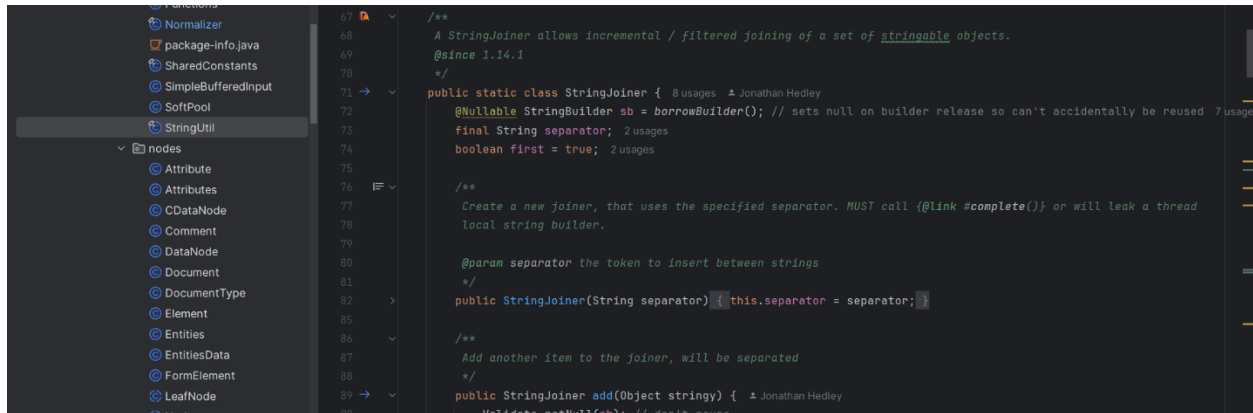


Figure 10: Unnecessary Abstraction(True Positive)

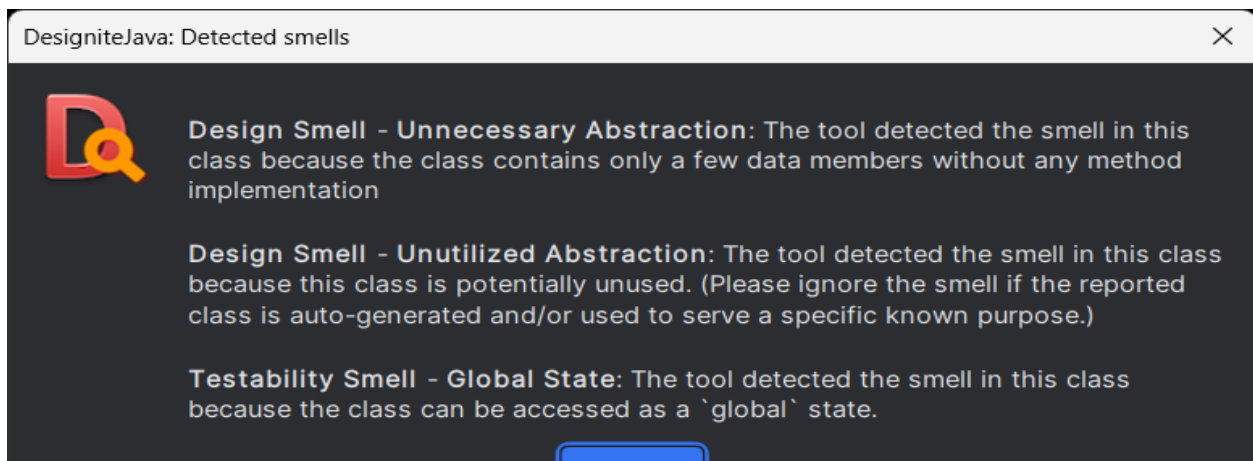


Figure 11: Tool Detected Design Smell Unnecessary Abstraction

Explanation

The StringJoiner class is identified as an Unnecessary Abstraction so, it a True Positive design smell. This class introduced an additional layer of complexity without providing significant benefits, as its functionality could be directly incorporated into the existing join methods. The abstraction was unnecessary because it only acted as a wrapper around a StringBuilder, which is already an efficient and widely used approach for string manipulation. So, I will use the Pull Down Method refactoring technique.

2) False Positive

2. Smell: Unutilized Abstraction

Location: src/main/java/org/jsoup/examples/Wikipedia.java

Line: 16

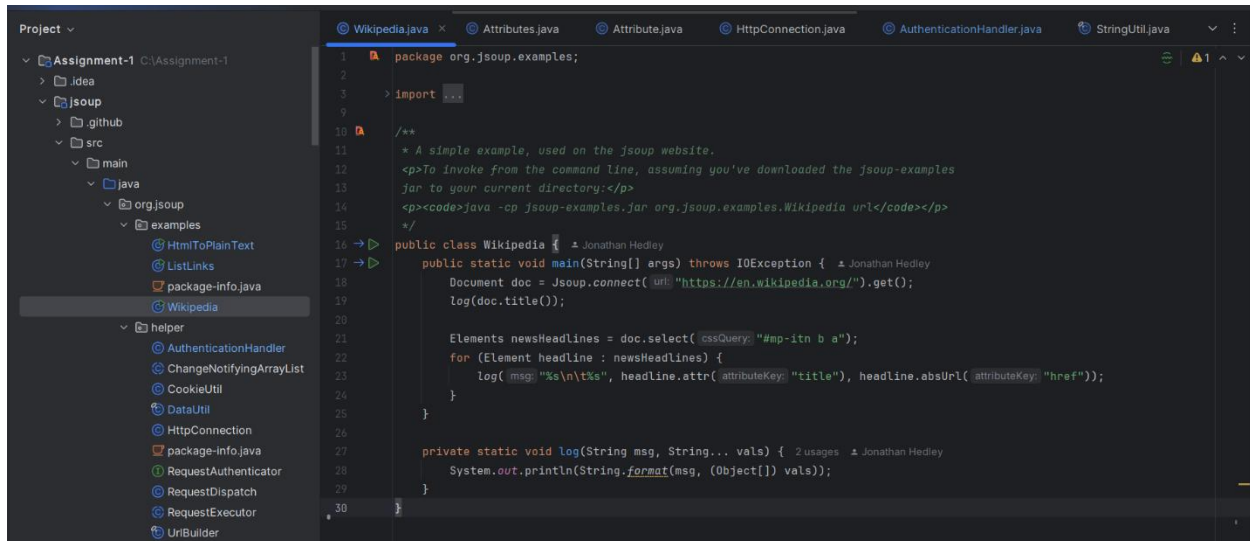


Figure 12: Design smell Unutilized Abstraction(False Positive)

Design Smell CSV file

11	java	org.jsoup	SerializationException	Unutilized Abstraction
12	java	org.jsoup	UnsupportedMimeTypeException	Unutilized Abstraction
13	java	org.jsoup.examples	Wikipedia	Unutilized Abstraction

Figure 13: Design smell CSV file

Explanation

The tool has incorrectly detected the main method in the Wikipedia class as an Unutilized Abstraction, which is a False Positive design smell. The main method serves as the entry point for executing this program, making it an essential part of the application's structure. Since this class is meant to be an executable example, the main method is required to initiate the process of fetching and displaying Wikipedia headlines. The likely reason for this false detection is that the designated tool may have interpreted the Wikipedia class as an abstraction that is not being utilized elsewhere in the code. However, this class is not meant to be used as an abstraction but rather as a self-contained executable.

3) False Negative

3. Smell: Deficient encapsulation

Location: src/main/java/org/jsoup/examples/ListLinks.java

Line: 18

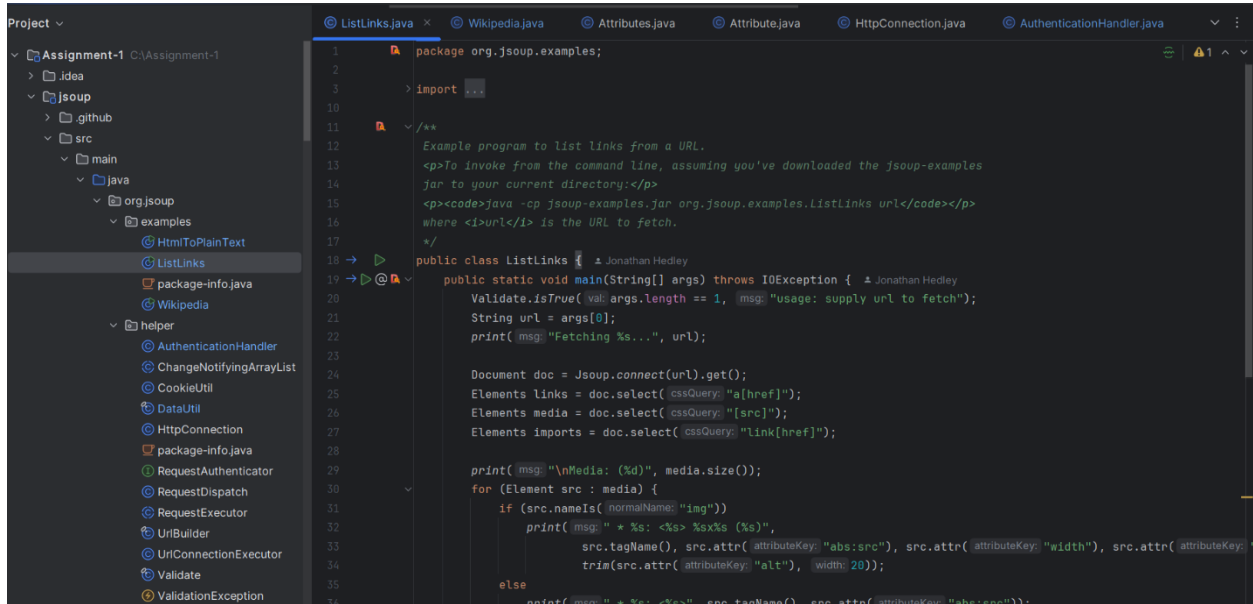


Figure 14: Design smell Deficient Encapsulation(False Negative)

Explanation

This is a false negative because the tool failed to detect a deficient encapsulation. The implementation of ListLinks directly handled HTML parsing, link extraction, and output formatting, leading to tight coupling between different responsibilities. Ideally, these concerns should have been encapsulated within separate classes to improve modularity and reduce dependencies. However, most static analysis tools primarily detect deficient encapsulation when fields are improperly accessed (e.g., using public fields instead of private ones) rather than when methods are misplaced within a class. Since this issue was related to method-level encapsulation rather than direct field access, the tool failed to flag it as a design smell.

3) Task 2

Set I – Refactoring of Implementation Smells using different techniques.

- 1) **Refactor name** – Decompose Conditional, Extract method, Introduce Explaining Variable

Location: src/main/java/org/jsoup/nodes/Attribute.java

Method: isValidHtmlKey

Line: 214

Before Refactoring Github Link:

<https://github.com/jhy/jsoup/blob/01b3900107114f441860ee3d03aba03caef42804/src/main/java/org/jsoup/nodes/Attribute.java>

After Refactoring Github Link:

<https://github.com/jhy/jsoup/commit/839eb0598f81fa23d047caaa4f41d1ac3837cec8>

Explanation

I refactored the isValidHtmlKey method to address the complex conditional implementation smell by decomposing the conditional logic, extracting a method, and introducing an explaining variable. Instead of checking multiple conditions inline, I introduced a helper method isInvalidCharacter, which improves readability by clearly defining what makes a character invalid. Additionally, I replaced the for loop with an enhanced for-loop using toCharArray(), making the code cleaner and more maintainable.

- 2) **Refactor name** – Decompose conditional and Introduce explaining variable

Location - src/main/java/org/jsoup/helper/DataUtil.java

Line – 238

Before Refactoring Github Link:

<https://github.com/jhy/jsoup/blob/01b3900107114f441860ee3d03aba03caef42804/src/main/java/org/jsoup/helper/DataUtil.java>

After Refactoring Github Link:

<https://github.com/jhy/jsoup/commit/b061b602214bd24a0fc53cd02ce794b4ad68b411>

Explanation

I refactored the detectCharset method by decomposing conditional logic and introducing explaining variables. The method was lengthy, had deeply nested conditionals, and handled multiple responsibilities, making it difficult to read and maintain. To improve readability, I extracted several helper methods. The getCharsetFromBom method isolates BOM-based charset detection, while parseInitialDocument handles the initial document parsing logic. Additionally, I separated charset detection logic into

detectCharsetFromDocument, which further delegates to findCharsetInMeta for detecting charset from <meta> tags and findCharsetInXmlDeclaration for extracting charset from XML declarations. This modular structure makes it easier to understand and modify each step of the process. I also introduced explaining variables, such as CHARSET_ERROR_MESSAGE, to replace inline strings and improve clarity. By simplifying conditional logic and improving code modularity.

3) **Refactor name:** Extract Method

Location: src/main/java/org/jsoup/helper/Authentication.java

Line: 14

Before Refactoring Github Link:

<https://github.com/jhy/jsoup/blob/01b3900107114f441860ee3d03aba03caef42804/src/main/java/org/jsoup/helper/AuthenticationHandler.java>

After Refactoring Github Link:

<https://github.com/jhy/jsoup/commit/1eebcc90e17b7b3fcb8630f939002c1aa14cb90d>

Explanation

I refactored the getPasswordAuthentication method to improve readability and maintainability by extracting a new helper method, shouldAuthenticate. This method encapsulates authentication condition checks, reducing complexity in the main method. Additionally, I fixed redundant code (such as checking delegate == null twice) and improved clarity by renaming variables and reformatting comments.

SET II - Refactoring of Design Smells using different techniques.

1) **Refactor name:** Pull down method

Location: src/main/java/org/jsoup/internal/StringUtil.java

Line: 41

Before Refactoring Github Link:

<https://github.com/jhy/jsoup/blob/01b3900107114f441860ee3d03aba03caef42804/src/main/java/org/jsoup/internal/StringUtil.java>

After Refactoring Github Link:

<https://github.com/jhy/jsoup/commit/55ebda6331a8eadc03a4803acb5c483c61b4fe77>

Explanation

I refactored the join method to eliminate unnecessary abstraction by reducing complexity. I simplified the flow by using a StringJoiner directly for concatenation, ensuring efficiency and clarity. I applied the Pull Down Method refactoring technique, where I moved logic into a helper class StringJoiner.

2) **Refactor name:** Move method, extract class, replace conditional with polymorphism

Location: src/main/java/org/jsoup/examples/ListLinks.java

Line: 18

Before Refactoring Github Link:

<https://github.com/jhy/jsoup/blob/01b3900107114f441860ee3d03aba03caef42804/src/main/java/org/jsoup/examples/ListLinks.java>

After Refactoring Github Link:

<https://github.com/jhy/jsoup/commit/988791b099ea910437e9c0507eb55f51e7dae43b>

Explanation

I have done refactor the code by using move method, extract class, replace conditional with polymorphism. I moved logic related to link extraction into a separate LinkExtractor class. I replaced the conditional logic for handling different link types (media, imports, hyperlinks) with polymorphism. This was achieved by introducing a LinkType interface and specific implementations (like MediaLink, ImportLink, and HrefLink) for different link categories. I also encapsulated the link creation logic in a factory pattern through the LinkFactory interface to reduce the complexity of link extraction.

Build Successful After Refactoring

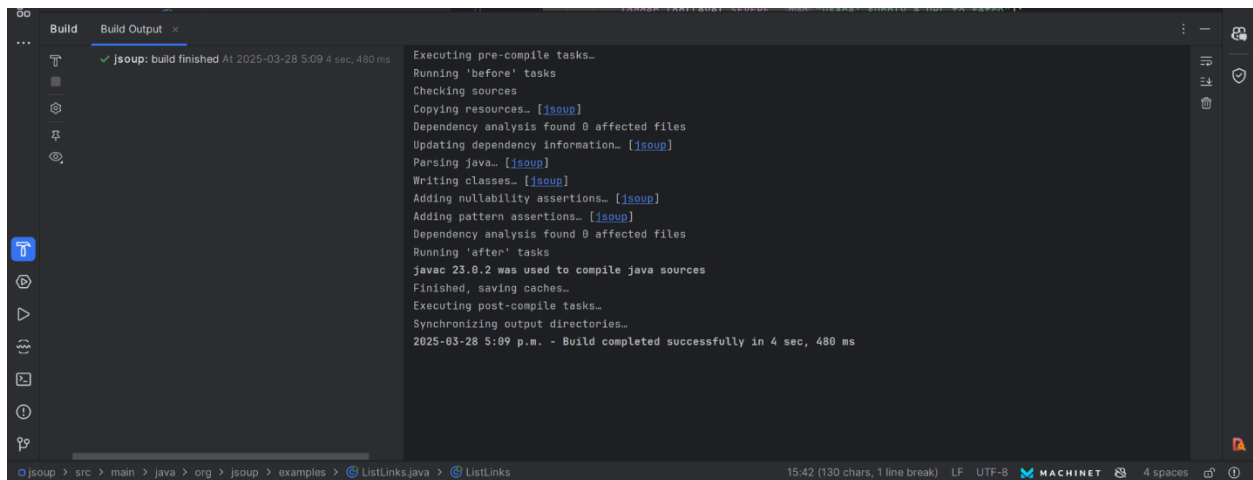


Figure 15: Build Successful After Refactoring

All Test cases Passed After Refactoring

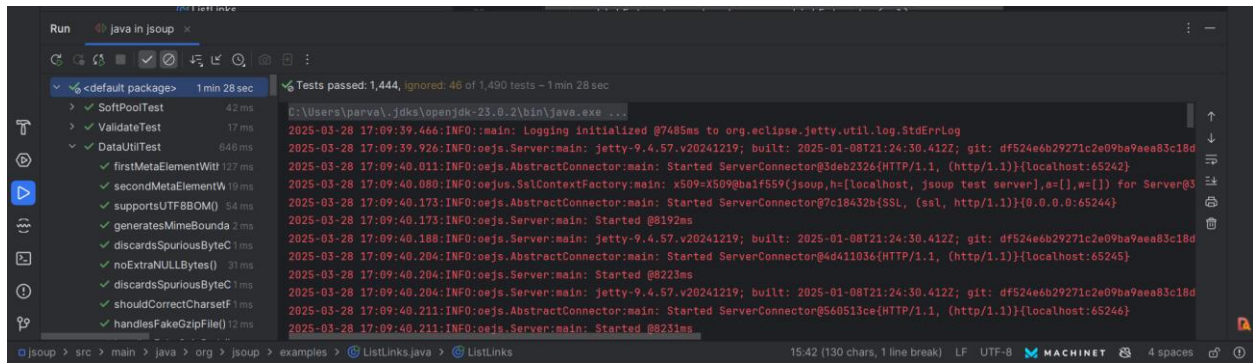


Figure 16: Test Cases Passed After Refactoring

4) Screenshot of Pull Request

The screenshot displays a GitHub Pull Request (PR) page for the repository `jhy/jsoup/pull/2292`. The page is titled "2612-Parva commented now" and shows a list of 25 commits by Parva Mineshbhai Patel, added 5 days ago. The commits are listed with their SHA hashes and descriptions. The sidebar on the right shows the PR status, including "No reviews", "Still in progress? Convert to draft", "Assignees: No one assigned", "Labels: None yet", "Projects: None yet", "Milestone: No milestone", "Development", "Successfully merging this pull request may close these issues.", "None yet", and "Notifications: Customize".

Conversation 0 Commits 25 Checks 0 Files changed 19 +681 -508

2612-Parva commented now

Complex conditional Use Decompose conditional, Extract method, Introduce Explaining Variable in Attribute.java
Deficient encapsulation by using move method, extract class, replace conditional with polymorphism in ListLink.java
Complex method by using Extract Method in AuthenticationHandler.java
Complex Method and Long Statement Decompose conditional and Introduce explaining variable into DataUtil.java
Unnecassry Abstraction by using Pull down method in StringUtil.java

Parva Mineshbhai Patel added 25 commits 5 days ago

- Design Smell False Positive Unutilized Abstraction 55a0751
- Design smell false negative Deficient Encapsulation 988791b
- False Positive Implementation smells Magic Number 8b7ebd7
- False negative Implementation smell Complex method 1eebcc9
- True Positive Implementation smell Complex Method and Long Statement a71c05d
- Design smell True Positive cyclically-dependent b9b9c36
- Remove Design smell True Positive cyclically-dependent bd62da2
- False Positive Design smells Unutilized Abstraction a4491c3
- False negative Design smell Deficient encapsulation 5b73138

Reviews
No reviews
Still in progress? [Convert to draft](#)

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.

None yet

Notifications [Customize](#)

[Unsubscribe](#)

Figure 15: Pull Request Raised

5) References

- [1] “DesigniteJava,” *Designite-tools.com*, 2025. Available: <https://www.designite-tools.com/products-dj>. [Accessed: Mar. 27, 2025]
- [2] J. Hedley, “jsoup: Java HTML Parser,” *GitHub* . Available: <https://github.com/jhy/jsoup>[Accessed: Mar. 27, 2025]
- [3] GitHub, “GitHub,” *GitHub*, 2025. Available: <https://github.com/> [Accessed: Mar. 27, 2025]